

Asthma in Pregnancy

Section 1: Case Summary

Scenario Title:	Asthma in Pregnancy
Keywords:	Asthma, pregnancy
Brief Description of Case:	A third trimester patient presents with a severe asthma exacerbation that deteriorates and requires BiPAP/ICU admission. The learner will need to identify severe asthma and manage accordingly accounting for differences in physiology during pregnancy.

Goals and Objectives	
Educational Goal:	Work through the steps of managing a progressively severe asthma exacerbation in pregnancy
Objectives: (Medical and CRM)	Medical: - Recognize physiologic changes in pregnancy (e.g. normal PCO₂ in pregnancy is 32mmHg) - Initiate FHR monitoring during medical emergency in pregnancy - Demonstrate stepwise pharmacological/respiratory support and treatments for severe asthma exacerbation in pregnancy CRM: - Early recognition of critically ill pregnant patient and the involvement of appropriate consultants - Communicate effectively with team to manage patient with multiple priorities
EPAs Assessed:	C7 - Managing urgent and emergent presentations for pregnant and post-partum patients

Learners, Setting and Personnel					
Target Learners:	☒ Junior Learners		☒ Senior Learners		☐ Staff
	☐ Physicians		☐ Nurses	☐ RTs	☐ Inter-professional
	☐ Other Learners:				
Location:	☒ Sim Lab		☐ In Situ		☐ Other:
Recommended Number of Facilitators:	Instructors: 2				
	Sim Actors: 0				
	Sim Techs: 1				

Scenario Development	
Date of Development:	April 26, 2023
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Last Revision Date:	
Revised By:	
Version Number:	



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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Janet			Age: 30	Gender: F	Weight: 80kg
Presenting complaint: Shortness of breath					
Temp: 36.9	HR: 95	BP: 99/59	RR: 32	O ₂ Sat: 85%	FiO ₂ : Room Air
Cap glucose: 7 mmol/L			GCS: 15 (E4 V5 M6)		
Triage note: 30 y/o F G1P0 at 35 weeks GA presents with shortness of breath increasing over the last few days. She has a cough and ran out of salbutamol overnight which is why she came to the ED this morning.					
Allergies: None					
Past Medical History: G1P0 Asthma: triggered by viral infections, irritants. Intubated x1 for exacerbation at age 11, ED visit and oral steroid 1 year ago. Previously followed by Family Physician but currently lost to follow up due to moving cities.			Current Medications: Salbutamol 100mcg inhaled as needed (dispensed 2 in the last month) Budesonide (ran out last month, has not renewed) Prenatal vitamins		

Section 2B: Extra Patient Information

A. Further History	
<i>Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, sim actors, SP, etc.)?</i>	
Patient has a history of 7 days of dry cough and had 3 days of fever that began 7 days ago. Fever has resolved but nonproductive cough is ongoing. Over the past 4-5 days she has been wheezing and short of breath. She was using her salbutamol 2 puffs every 4 hours initially, but since yesterday about every 1-2 hours. She feels that her wheezing and shortness of breath is getting worse. There is no chest pain or pleuritic symptoms.	
This is her first pregnancy, she is at 35 weeks GA. She was followed by a family health team in another province until 22 weeks GA when she moved here. Her dating and anatomy scans showed a single live intrauterine pregnancy with no anomalies. There is no abdominal pain, contractions, vaginal bleeding or rupture of membranes. She is feeling regular fetal movements.	
There is no history of leg swelling, immobilization, trauma or surgery. Her family also had URTI symptoms and fever. There is no travel history and her vaccines are up to date.	
B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
Cardio: S1S2 no murmurs, no peripheral edema	Neuro: GCS 15, A0x3, strong x 4
Resp: Increased WOB with accessory muscle use. Decreased air entry bilaterally. Inspiratory and expiratory wheezes throughout. No crackles.	Head & Neck: No angioedema, no neck swelling, no dysphonia/stridor



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Abdo: Gravid abdomen, uterus palpable between umbilicus and xyphoid process	MSK/skin: No abnormalities
Other: Nil	

Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin- Adult mannequin with T-shirt and pillow to simulate gravid abdomen
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Monitor Fetal heart rate monitor Bipap mask and machine Nasal prongs, face mask
C. Required Medications
Salbutamol Ipratropium Methylprednisolone/Prednisone Magnesium Sulfate Ketamine
D. Moulage
T-shirt and pillow to simulate gravid abdomen
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed
☒ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> A – nil B – inspiratory and expiratory wheezes, decreased air entry C – Gravid uterus, abdomen soft nontender D – nil E – nil

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Section 4: Sim Actor and Standardized Patients

Sim Actor and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Patient	Speaks in 2-3 word sentences due to shortness of breath. "I ran out...of my puffers" "Is my baby...going to be okay?"
Facilitator/ confederate	Can verbalize changes in patient's exam through the case if specific findings of work of breathing are not asked for, to provide realistic intervals between reassessment and status changes (script in patient status column).

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Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: sinus HR: 95 BP: 99/59 RR: 32 O ₂ SAT: 85% T: 36.9 °C GCS: 15	<i>Is the patient alert? In distress? Seizing? What symptoms do they currently have?</i> Alert, moderate respiratory distress, audible wheezing	<u>Expected Learner Actions</u> ☐ Place patient on monitors ☐ Ask for FHR monitoring ☐ Target 95% with O ₂ via nrb ☐ Take brief history including obstetrical history ☐ Performs primary survey ☐ Places the patient in lateral or seated position ☐ Initiates salbutamol/ipratropium inhalers/nebs ☐ Calls for RT and FEV1/PEF measurement ☐ Initiate IV access	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Does not apply supplemental O ₂ → Sat drops to 80% - Applies O ₂ → Sats increase to 93% - lateral or seated position (or wedge towards left), BP increases to 106/68 <u>Triggers</u> <i>For progression to next state</i> - Initiates salbutamol/ipratropium inhalers or nebs → Phase 2	- Does not ask for FHR monitoring → Patient asks “is my baby ok” to prompt -If asks for FEV1/PEF: 40% of predicted maximum
2. Deterioration Rhythm: sinus HR: 112 BP: 106/68 RR: 40-50 O ₂ SAT: 93% T: 36.9 °C GCS: 14 FHR: 160	Alert, severe respiratory distress, audible wheezing, work of breathing increases “The patient has completed salbutamol and ipratropium nebs back to back as per your order. She begins to look fatigued and has	<u>Expected Learner Actions</u> ☐ Orders labs including venous blood gas ☐ ECG ☐ CXR ☐ ICU, OB consult ☐ Initiate BiPAP ☐ Give methylprednisolone IV	<u>Modifiers</u> - If epinephrine ordered, increase FHR to 170 <u>Triggers</u> - Initiates BiPAP and gives methylprednisolone → Phase 3	Settings for BiPAP: PEEP at 5 cmH ₂ O (low) iPAP at 7-15 cmH ₂ O, adjust to target TV of 5-6mL/kg High inspiratory flow rate Low I:E ratio (e.g. 1:5) Prolonged expiratory time Initial dose methylpred: 60-80mg IV q6h (then taper) If not initiating BiPAP, nurse or RT can provide prompt “should we consider NIPPV in this patient?”



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	severe work of breathing.”			
3. Stabilization and handover Rhythm: sinus HR: 105 BP: 106/68 RR: 27 O ₂ SAT: 96% T: 36.9 °C GCS: 15 FHR: 160	Alert, moderate respiratory distress, wheezing on auscultation with improved air entry, work of breathing improved “1h after the patient has initiated BiPAP, there is ongoing wheezing and moderate WOB. The air entry has improved.”	<u>Expected Learner Actions</u> ☐ Interprets CXR, ECG, labs (resp acidosis) ☐ Considers magnesium, IV salbutamol or terbutaline ☐ Handover to OB/ICU	<u>Modifiers</u> - Orders magnesium: decrease in respiratory distress, improved air entry, work of breathing improved - Orders magnesium and asks for repeat vitals: HR: 100 BP: 106/68 RR: 25 O ₂ SAT: 98% T: 36.9 °C FHR: 150 - If epinephrine ordered, increase FHR to 170 <u>Triggers</u> - Orders magnesium - Consider terbutaline, IV salbutamol - Handover to OB/ICU after one adjunct med given + reassessment	<u>Dosing:</u> terbutaline 0.25 mg SC q20 minutes up to 3 doses magnesium sulfate 2g IV over 20 minutes salbutamol infusion: 5 mcg/minute; may increase up to 10 to 20 mcg/minute at 15- to 30-minute intervals, if needed.



Simulation Scenario Template

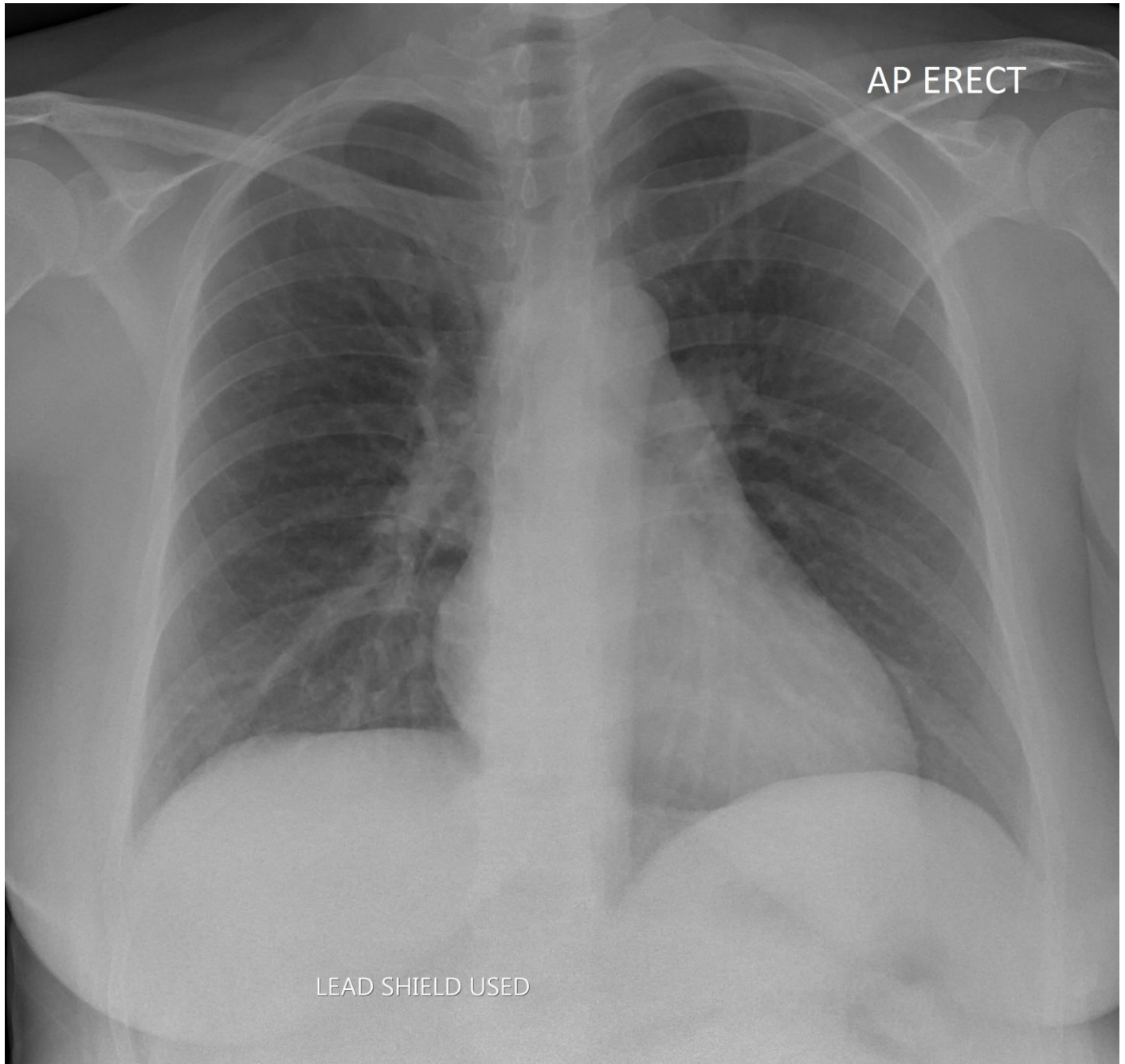
Appendix A: Laboratory Results

<u>CBC</u> WBC 14.1 Hgb 118 Plt 150 <u>Chemistry</u> Na 140 K 4.8 Cl 100 HCO ₃ 19 AG 10 Urea 3 Cr 75 Glucose 7 Ca 2.32 Mg 1.00 Albumin 40 TSH 2.10 <u>VBG</u> pH 7.32 pCO ₂ 40mmHg pO ₂ 69mmHg HCO ₃ 19 Lactate 3.0	<u>Cardiac/Coags</u> hs-Trop 5 D-dimer 490 INR 1.0 aPTT 30
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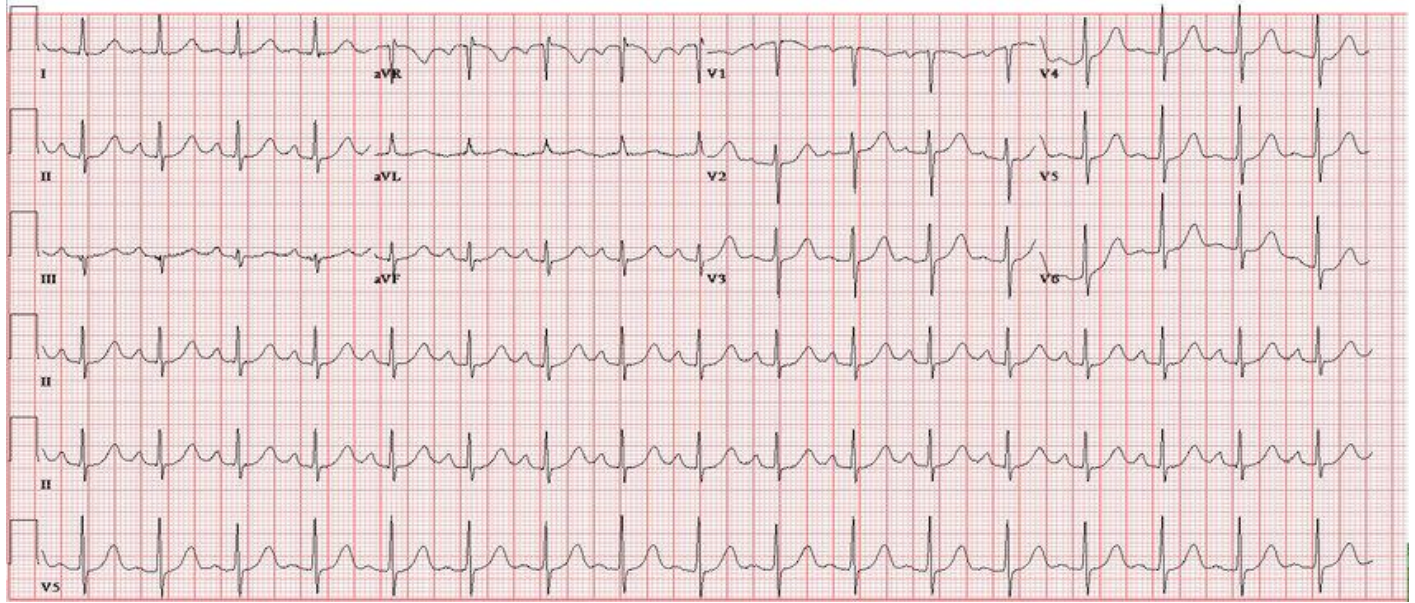
Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!



Case courtesy of Henry Knipe, [Radiopaedia.org](https://radiopaedia.org). From the case rID: [36714](https://radiopaedia.org/cases/36714)

Simulation Scenario Template



Patel, Nirav & Negi, Smita & Anand, Aashish & Rao, Anantha. (2016). The New Zealand medical journal. 129. 80-83.

Simulation Scenario Template

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

CXR interpretation: Prominence to the vascular markings (normal in pregnancy). Normal cardiomedial contour. Lungs and pleural spaces are clear.

ECG interpretation: Sinus tachycardia at 103 bpm

PE is a reasonable consideration in the DDX. YEARS algorithm in pregnant patient can rule out PE in this patient with no hemoptysis, no signs/sx of DVT and a reasonable alternative diagnosis.

Objectives:

- Recognize physiologic changes in pregnancy (e.g. normal PCO₂ in pregnancy is 32mmHg)
- Initiate FHR monitoring during medical emergency in pregnancy
- Demonstrate stepwise pharmacological/respiratory support and treatments for severe asthma exacerbation in pregnancy
- Stepwise pharmacological/respiratory support treatments for severe asthma exacerbation in pregnancy: 1 – Bronchodilators 2 – Ipratropium 3 – systemic corticosteroids 4 – Adjunct therapies (Magnesium sulfate, terbutaline)
- Avoid epinephrine due to concern for uterine vasoconstriction

References

1. Rosen's emergency medicine : concepts and clinical practice. Chapter 174: Medical Emergencies During Pregnancy (2022). St. Louis :Mosby,
2. Awad, N. Fleetham, J. Asthma in Pregnancy. Journal of Obstetrics and Gynecology Canada. Volume 17, Issue 1, p27-31, January 1995.
3. Schatz, W., & Weinberger, S. (2022). Management of asthma in pregnancy. In Dieffenbach (Ed.), UpToDate.

